

SM400 Brushless AC Servo Motor

SKU / Model Number: SM400-3K5-B
Manufacturer: DriveForce Automation
Category: Motors | ETIM Code: EC001851

PRODUCT OVERVIEW

The SM400 is a high-performance brushless AC servo motor with integrated encoder, designed for precision motion control in CNC machines, robotics, and packaging equipment. Features rare-earth magnets and low cogging torque design.

TECHNICAL SPECIFICATIONS

Rated Power	3.5 kW
Rated Torque	11.2 Nm
Peak Torque	33.6 Nm
Rated Speed	3000 RPM
Max Speed	5000 RPM
Torque Constant	1.42 Nm/A
Rotor Inertia	7.5 x 10 ⁻⁴ kg.m ²

ELECTRICAL RATINGS

Rated Voltage	400 V AC, 3-phase
Rated Current	8.5 A
Peak Current	25.5 A
Back-EMF Constant	175 V/kRPM
Winding Resistance	1.2 ohm (phase-phase)

DIMENSIONS & MECHANICAL

Frame Size	130 mm
Body Length	185 mm
Shaft Diameter	22 mm
Shaft Length	50 mm
Weight	12.4 kg

OPERATING CONDITIONS

Ambient Temperature	0 to 40 deg C
Insulation Class	Class F (155 deg C)
Cooling	Natural convection
Vibration	< 5 mm/s (IEC 60034-14)
IP Rating	IP65

MATERIALS OF CONSTRUCTION

Stator Core	Silicon Steel Laminations
Magnets	NdFeB (N42SH)
Shaft	42CrMo4 alloy steel
Housing	Aluminum alloy (die-cast)
Windings	Class H copper wire

PERFORMANCE DATA / SELECTION TABLE

Model	Power (kW)	Torque (Nm)	Speed (RPM)	Weight (kg)
SM400-1K0	1.0	3.2	3000	5.2
SM400-2K0	2.0	6.4	3000	8.1
SM400-3K5	3.5	11.2	3000	12.4
SM400-5K0	5.0	16.0	3000	16.8
SM400-7K5	7.5	24.0	3000	22.0

CERTIFICATIONS & APPROVALS

CE | UL | CSA | CCC | RoHS | REACH

Note: Specific certification documents and test reports are available upon request. All products are manufactured under ISO 9001:2015 quality management system.

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